MLB Sticky Stuff Analysis

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Origination

Introduction

Topic:

- In June 2021, the MLB implemented new rules that regulated pitchers usages of foreign substances “sticky stuff”
- This ban was aimed at tilting the advantage to hitters, consequently making the games more “interesting.”

Importance:

- Sticky stuff was widely used throughout the league, to increase command, spin rate, and velocity
- Many believed, it gave pitchers an unfair advantage, helping them strike out more batters, while giving up less walks and hits, ultimately leading to pitcher statistical dominance

Introduction

Data Sources

- MLB Stats API
- Statcast / Baseball Savant

Target Audience of our Analysis

- MLB League Officials - To see how effective these rule changes were.
- Pitchers - To see how their “stuff” and statistics were affected by these changes.
- Scouting Departments - To make a more accurate analysis when scouting pitchers, on how they may perform with the new rule changes.

Motivation

- As baseball fans and savants, we were genuinely interested to see the effect of the ban different pitchers, and how pitcher performance statistics like strikeouts, ERA, and hits/home runs allowed were affected.
- Do the rule changes make baseball shift baseball to a more “interesting” offensive game or did the ban solely affect optics?
- As a pitcher myself, who has had a habit of using sticky stuff from time to time, I wondered how much did it truly affect my pitch metrics (velocity, spin rate, movement) and performance. Thus, our analysis reveals whether I should continue using sticky substances at the college level.

Guiding Questions

1. Did league-wide strikeout rates and pitching dominance decline after the crackdown?

2. Which individual pitchers showed the largest performance drops, and do those names match the public narrative?

3. Are the observed changes statistically significant, even after controlling for injuries and aging?

4. Does Statcast spin rate data confirm the mechanism — did pitchers actually lose spin after enforcement?

Process

Tools & Methods

Data Collection

- MLB Stats API — season-level pitching stats (ERA, WHIP, K%, BB%, BAA, HR/9) for all pitchers, 2019–2023
- Baseball Savant (pybaseball) — 2.6M pitch-level records (spin rate, velocity, movement)
- MLB Transactions API — 60-day IL stints as an injury control variable

Python Analysis

- Cleaned and merged three datasets into one DataFrame; loaded into SQLite for SQL analysis
- Feature engineering: age buckets, pitch profile categories, interaction terms (spin × velocity), injury flags, IQR outlier capping
- EDA: trend lines, KDE distributions, correlation heatmaps, annotated scatter plots
- Paired and independent t-tests with age-stratified and healthy-only subgroups
- Linear regression (predict K% change) and logistic regression (classify most-affected pitchers)

Tools & Methods

SQL Analysis (15 queries)

- 13 JOINs linking pitching stats, Statcast data, and injury records
- 7 window function types: RANK, DENSE_RANK, ROW_NUMBER, LAG, LEAD, NTILE, running AVG
- ROLLUP and CUBE (via UNION ALL) for multi-dimensional summaries
- Correlated and non-correlated subqueries for pitcher-vs-league comparisons

Results

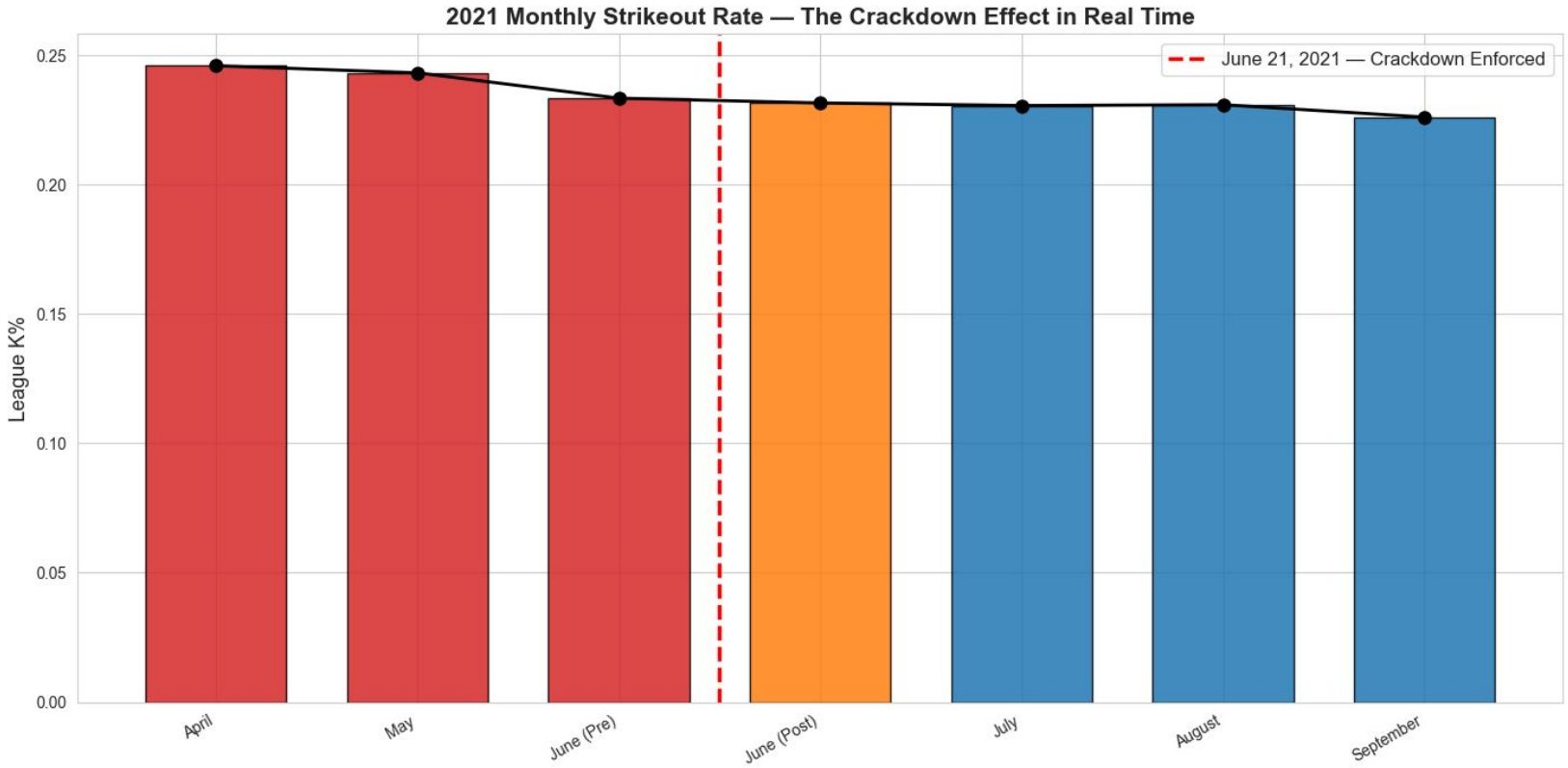
Did strikeout rates and pitcher dominance fall?

Yes, but...

- We did see strikeout rates fall after the crackdown date with clear inflection point in 2021
- However, evidence is strongest when same pitchers are compared before and after, rather than when pools of pitchers are compared across seasons
  - Notably, the independent t-test comparing K% across all pitchers pre vs. post was **not** significant ($p = 0.515$), while the paired t-test on the same pitchers was highly significant ($p < 0.001$), confirming that roster turnover between seasons masks the effect
  - Among the 175 pitchers who qualified in both eras, average K% dropped from **25.05% to 23.34%** (paired $t = 4.40$, $p < 0.001$)
- Overall, crackdown lowered ability to get strikeouts, but effect is clearer when controlling for pitcher

2021 Monthly K% — The Inflection Point

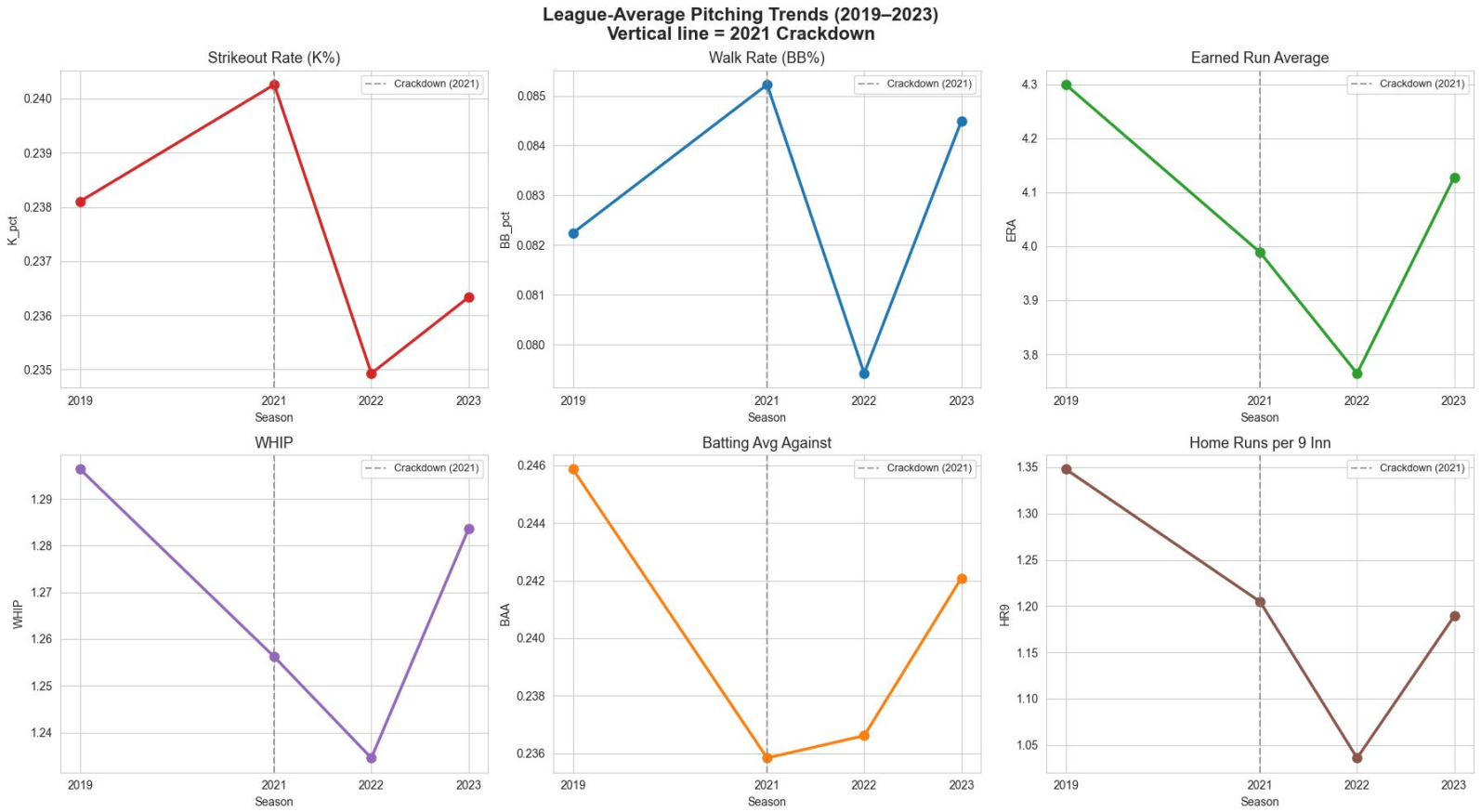
- League-wide strikeout rate visibly drops after the June 21, 2021 enforcement date
- Supports a crackdown-driven cause rather than a gradual seasonal trend



League Trend Analysis

- **K% (Strikeout Rate):** Declined post-crackdown (23.8% → 23.6%), consistent with pitchers losing spin-based swing-and-miss advantage
- **BB% (Walk Rate):** Remained stable across the entire period — pitchers didn't lose command, just whiff ability. This supports a spin-specific effect rather than a general decline in pitching quality
- **ERA:** Actually *improved* post-crackdown (4.30 → 3.95), likely driven by confounding factors like ball composition changes and roster trends rather than the crackdown itself
- **WHIP:** Also improved slightly (1.30 → 1.26), same caveat — other concurrent MLB changes muddy this signal
- **BAA (Batting Average Against):** Shows more complex patterns influenced by factors beyond the crackdown (shifts ban, ball changes)
- **HR/9 (Home Runs per 9 Innings):** Also shows complex patterns — home run rates are heavily influenced by ball composition, which MLB changed multiple times during this period

League Average Statistical Trend Chart (K%, BB%, ERA, WHIP, BAA, HR/9)



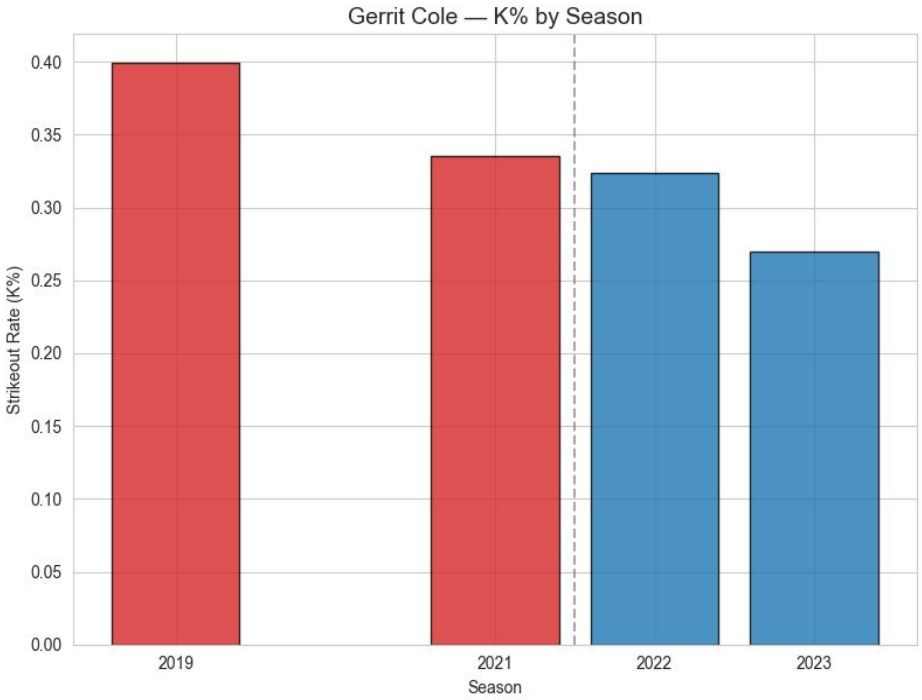
Which Pitchers Had The Biggest Drops in Performance? Does That Match The Public Narrative?

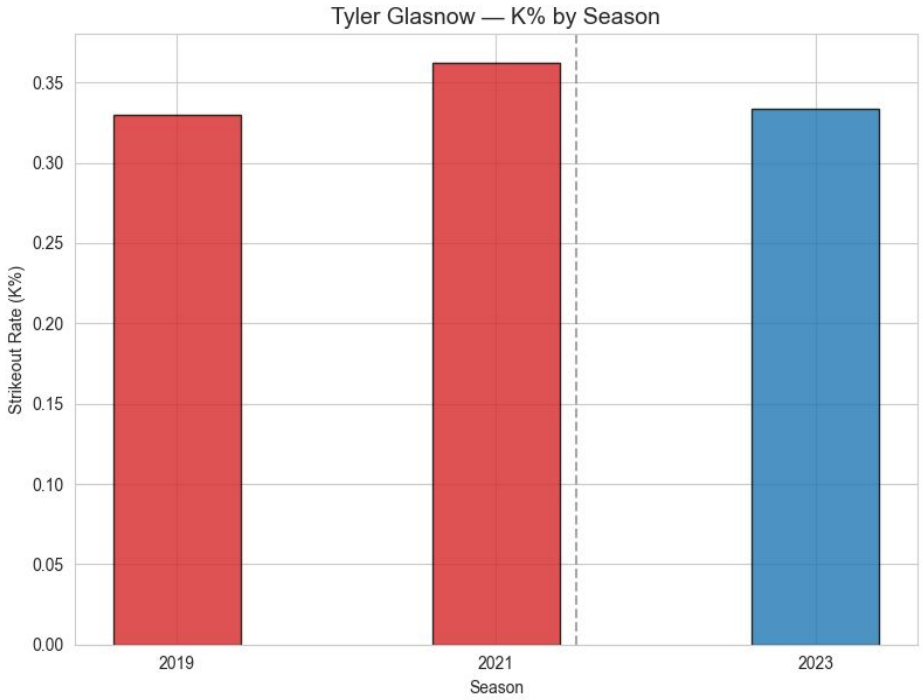
Yes.

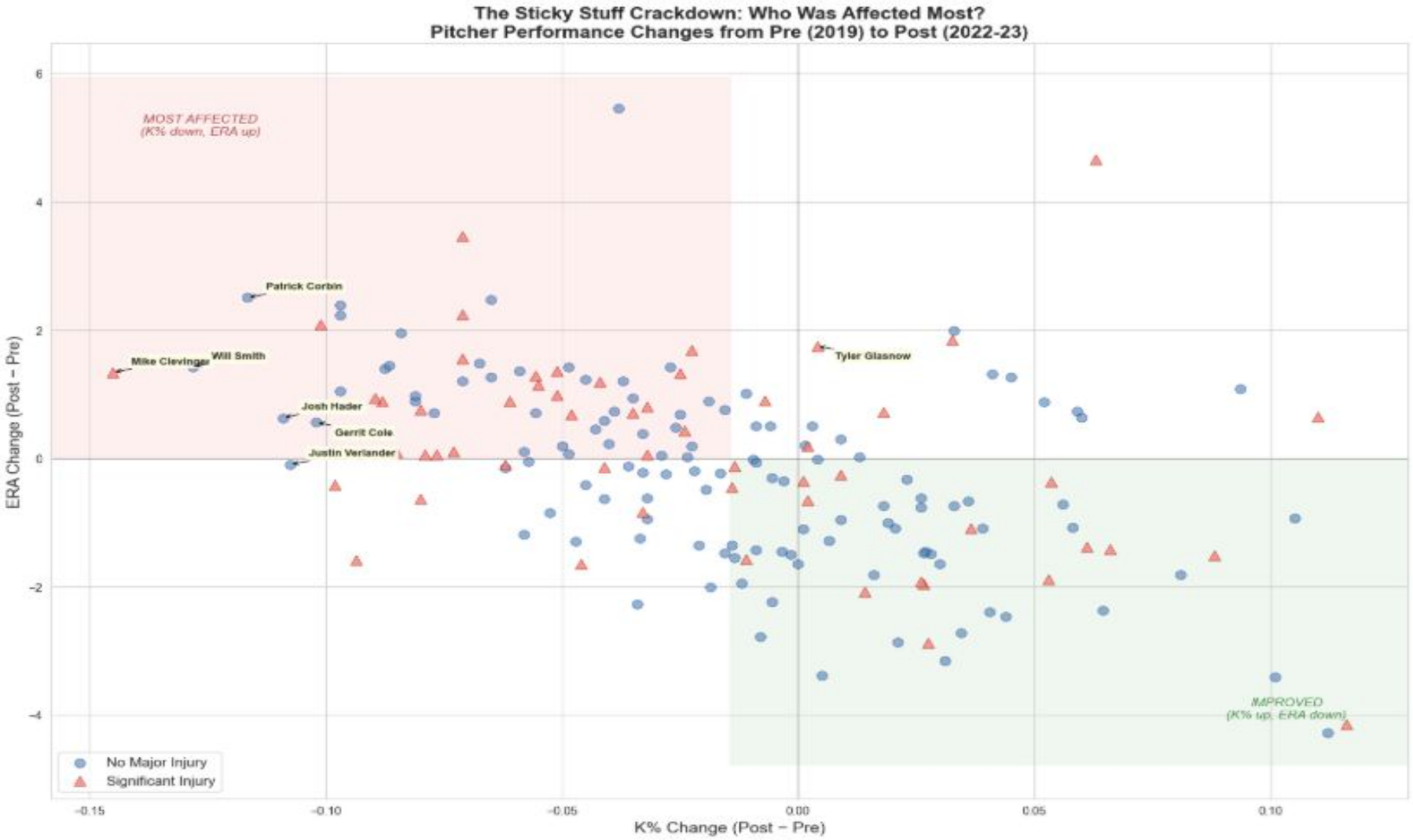
Pitchers with the largest declines performance, were publicly regarded as as users of ‘sticky stuff’

- **Gerrit Cole:** noticeable decline in strikeout percentage
- **Tyler Glasnow:** large increase in ERA, strikeout percentage roughly the same, although increase in ERA could’ve been due to UCL injury suffered post-crackdown

Visualization of Gerrit Cole & Tyler Glasnow K% by Season






The Sticky Stuff Crackdown: Who Was Affected Most?
Pitcher Performance Changes from Pre (2019) to Post (2022-23)
MOST AFFECTED
(K% down, ERA up)
IMPROVED
(K% up, ERA down)
Patrick Corbin
Mike Clevinger
Will Smith
Josh Hader
Gerrit Cole
Justin Verlander
Tyler Glasnow
No Major Injury
Significant Injury
ERA Change (Post − Pre)
K% Change (Post − Pre)
6
4
2
0
−2
−4
−0.15
−0.10
−0.05
0.00
0.05
0.10

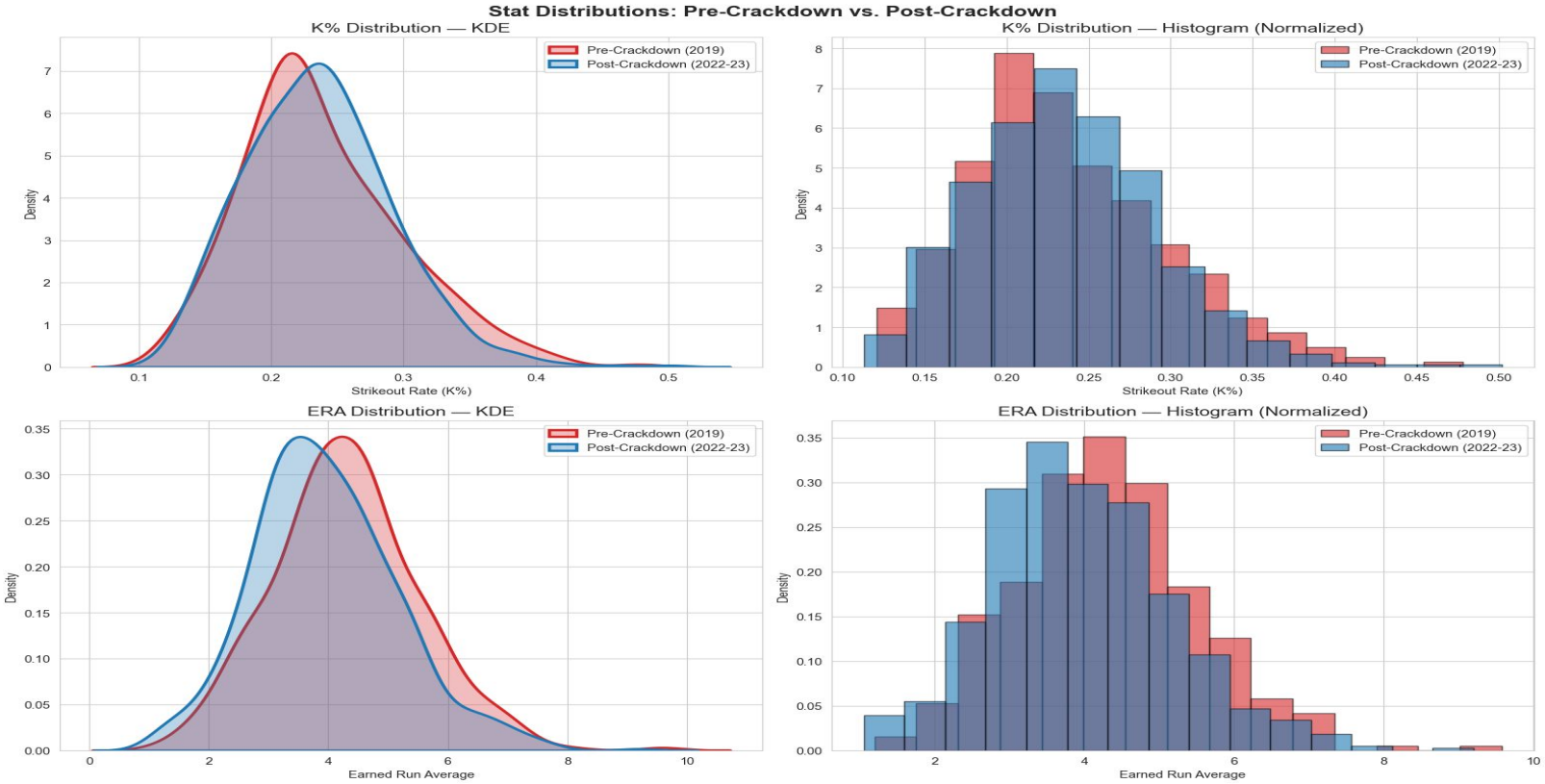
Are Observed Changes Statistically Significant, Even After Controlling For Injuries and Aging?

Yes.

The decline in strikeout rates is statistically significant when controlling for injuries and aging:

- When restricting to only healthy pitchers, decline is still statistically significant
  - Strikeout rate drop with age-group:
    - Pitchers under 30: $p = .0056$
    - Pitchers over 30: $p < .0001$
      - Both age groups showed significant decline
- As aforementioned, Among the 175 pitchers who qualified in both eras, average K% dropped from **25.05% to 23.34%** (paired $t = 4.40$, $p < 0.001$)
- Evidence strongly supports real decline in strikeout ability post crackdown

K% & ERA Distribution Plots (Pre vs. Post Overlay)



Does Statcast Spin Rate Support Our Hypothesis?

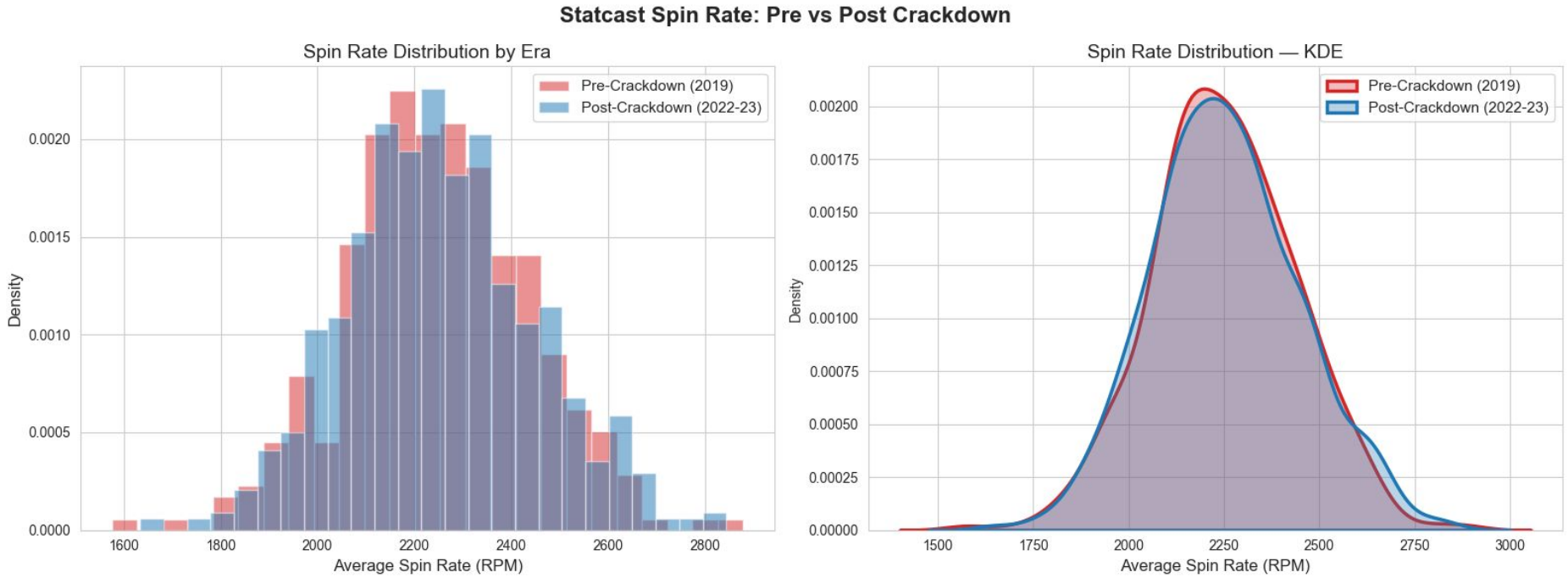
Yes.

Adding in spin rate data to our analysis showed:

- Spin rates declined after crackdown
- Regression found that spin rate change was a **statistically significant predictor** of strikeout rate change ($p < 0.001$), though the model explains roughly 15–30% of variance — consistent with baseball's inherent noise

Overall, adding spin rate data allowed us to hypothesize that foreign substances led to higher pitcher spin rate and thus more strikeouts. So, after the crackdown pitchers had less grip and thus had lower strikeout percentage.

Spin rate distribution plots (pre vs. post)



Logistic Regression – Who Was “Significantly Affected”

- We built a model to predict which pitchers were most hurt by the crackdown, based on their spin rate, velocity, injury history, and other stats
- Out of 35 pitchers tested, the model got the **right call 23 times** (65.7% overall)
- When the model flagged a pitcher as affected, it was **right 3 out of 4 times** (73.3%), rarely raising false alarms
- The tradeoff: it was conservative, missing some truly affected pitchers in the process (only caught **about 6 out of 10** of them (57.9%)
- Overall confidence score: **0.74 out of 1.00**
    - Solid but not perfect predictor, makes sense given how many factors influence pitcher performance

Limitations

- 2020 season excluded due to COVID-shortened 60-game season, creating a gap in the time series
- Roster turnover between seasons means pre and post groups aren't identical — paired tests on the same pitchers are more reliable than cross-season comparisons
- Linear regression R-squared of 15–30% means the model captures the crackdown signal but most of pitcher performance variance comes from factors we don't measure
- We cannot directly observe sticky stuff usage — we infer it through spin rate changes, which is an indirect proxy

Key Takeaways

- The crackdown worked
  - strikeout rates declined significantly post-2021, and the effect holds after controlling for injuries and aging
- The most affected pitchers match the public narrative
  - names like Gerrit Cole appear at the top of decline lists
- Spin rate is the mechanism
  - pitchers who lost the most spin lost the most strikeouts, confirming that foreign substances were artificially boosting performance
- The effect is real but modest
  - baseball has many variables, and the crackdown was one factor among many

Thank you!

Appendix:

Process - Python

Data Collection

- **MLB Stats API** — requests.get() to pull season-level pitching stats (ERA, WHIP, K%, BB%, BAA, HR/9, IP) for all pitchers, 2019–2023
- **Baseball Savant** — pybaseball.statcast() to pull 2.6 million pitch-level records (spin rate, velocity, horizontal/vertical movement)
- **Injury data** — MLB API transactions endpoint to flag pitchers with 60-day IL stints as a control variable

Data Cleaning & Preparation

- pd.to_numeric(errors="coerce") to handle non-numeric entries
- .isna().sum(), .duplicated(), .describe() to check for oddities and validate data quality
- pd.merge() to combine standard stats, advanced stats, Statcast aggregates, and injury flags into one DataFrame
- Loaded all three datasets into a **SQLite database** via df.to_sql() for SQL analysis

Process - Python

Feature Engineering

- df.apply() and df.map() to create age buckets and throwing hand features
- pd.get_dummies(drop_first=True) for one-hot encoding of pitch profile
- Interaction terms (spin × velocity), polynomial features ($spin^2$), and ratio features (K/BB ratio, spin per mph)
- IQR method for outlier detection and capping

Exploratory Data Analysis

- Line plots for league-wide trends by season
- KDE plots and histograms for distribution comparisons (pre vs. post crackdown)
- Correlation heatmaps by era (sns.heatmap)
- Pair plot of Statcast and performance variables (sns.pairplot)
- Three aggregate summary tables with display()
- Annotated scatter plot of K% change vs. ERA change with injury flags

Process - Python

Statistical Testing

- **Paired t-test** (stats.ttest_rel) — same pitchers, pre vs. post → K% decline significant ($p < 0.001$)
- **Independent t-test** (stats.ttest_ind) — league-wide ERA, WHIP, BAA comparisons
- Age-stratified and healthy-only subgroup tests to control for confounders

Modeling

- **Linear Regression** — predict K% change from Statcast + traditional features
  - train_test_split, StandardScaler, R^2, adjusted R^2, RMSE
  - statsmodels OLS for coefficient p-values
- **Logistic Regression** — classify pitchers as "significantly affected" (K% decline > median)
  - Accuracy, precision, recall, ROC AUC, confusion matrix

Process - SQL

Database: mlb_sticky_stuff.sqlite with 3 tables loaded via df.to_sql()

- pitching_stats — season-level traditional stats (K%, ERA, WHIP, BAA, era labels)
- statcast_agg — aggregated spin rate, velocity, movement per pitcher-season
- injuries — 60-day IL transactions

JOINs (13 queries)

- LEFT JOIN pitching stats with Statcast data to link performance to spin rate
- LEFT JOIN pitching stats with injury details to identify injury-related declines
- JOIN pre-crackdown and post-crackdown stats to compute individual pitcher changes

Process - SQL

Window Functions (8 queries, 7 function types)

- RANK() — rank pitchers by spin rate within each era
- DENSE_RANK() — compare spin rate vs. velocity rankings to identify suspicious gaps
- ROW_NUMBER() — deduplicate to each pitcher's best season per era
- LAG() — year-over-year spin rate changes (when did the drop happen?)
- LEAD() — next-season projections (who was on a collision course with enforcement?)
- NTILE(4) — divide pitchers into K% decline quartiles
- Running AVG() with ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW — cumulative spin rate trajectory

ROLLUP & CUBE (2 queries, via UNION ALL for SQLite)

- ROLLUP — average stats by era × injury status with subtotals and grand total
- CUBE — stats by era × pitch profile (fastball-heavy vs. offspeed-heavy) with all 2^2 grouping combinations

Subqueries (7 queries)

- Non-correlated — filter to pitchers above league-average spin rate using (SELECT AVG(avg_spin_rate) FROM statcast_agg)
- Correlated — compare Gerrit Cole and Tyler Glasnow's K% to the league average each season using (SELECT AVG(K_pct) FROM pitching_stats WHERE season = p.season)